

## PRESS RECORD

# COMPX341 - Advanced Software Engineering Design



THE UNIVERSITY OF  
**WAIKATO**  
*Te Whare Wānanga o Waikato*

# Before we start ...

- Guest speaker this Friday
- Please come along in person



Liam Clow  
DevOps Engineer  
Hamilton City Council

# DevOps – an Introduction

COMPX<sub>341</sub>

# Overview

- What is DevOps?
- What is SRE?
- DevOps/SRE and the Cloud

What is DevOps?

# What is DevOps?

- Dev
  - Development teams focus on the software product
- Ops
  - System administrator teams (sysadmin) focus on operations
  - i.e. Assembling/deploying software to run a service
  - i.e. Responding to events/updates/traffic/etc.
- DevOps:
  - Run Ops as a software engineering project
  - Design/build software to automate sysadmin/ops tasks
  - Mixed skills needed!

What is SRE?

# What is Site Reliability Engineering?

“By design, it is crucial that SRE teams are focused on engineering. Without constant engineering, operations load increases and teams will need more people just to keep pace with the workload. Eventually, a traditional ops-focused group scales linearly with service size: if the products supported by the service succeed, the operational load will grow with traffic. That means hiring more people to do the same tasks over and over again.” – *Benjamin Treynor Sloss, Google SRE Book*

# What is Site Reliability Engineering?

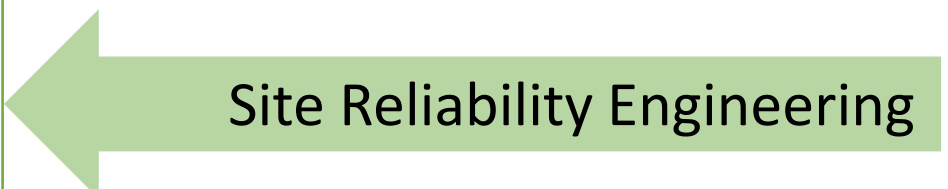
“To avoid this fate, the team tasked with managing a service needs to code or it will drown.”

“... the SRE team should end up with very little operational load and almost entirely engage in development tasks, because the service basically runs and repairs itself: we want systems that are **automatic**, not just **automated**. ”

– Benjamin Treynor Sloss, *Google SRE Book*

# What is Site Reliability Engineering?

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**Site Reliability Engineering**

# What is Site Reliability Engineering?

SRE is DevOps, with some caveats

- SRE is DevOps done by Software Engineers
- SRE is DevOps with a set of core tenets

# Tenets of SRE

Site Reliability Engineering has eight tenets:

1. Availability
2. Latency
3. Monitoring
4. Emergency response
5. Change management
6. Capacity planning
7. Efficiency
8. Performance

And SRE teams are responsible for upholding these tenets

# Tenet 1: availability

“Availability refers to the percentage of time that the infrastructure, system, or solution remains operational under normal circumstances in order to serve its intended purpose.” — <https://www.bmc.com/blogs/reliability-vs-availability/>

- SRE: 100% availability is the wrong target
- Instead:
  - What level of availability will the users be happy with?
  - What alternatives are available?
  - What happens to usage of the product at different availability levels?
- Can use remaining percentage as ‘error budget’
  - i.e. Error budget =  $1 - \text{availability}$ . E.g. 0.999 availability leaves 0.001 (or 0.1%) error budget
  - i.e. can spend the error budget on innovations and feature velocity

## Tenet 2: latency

- Humans add latency
- Imagine two systems
  - System A: relies heavily on human decisions and interventions
  - System B: able to avoid some human interventions
  - Even if System B experiences more failures than System A, it will have higher availability because it does not rely on human decisions and interventions as heavily, i.e. Avoids high levels of latency
- On-call playbooks
  - No system can avoid all human interventions
  - In these cases, SRE relies on on-call playbooks to help reduce latency
  - Playbooks are created by thinking and recording best practices ahead of time

# Tenet 3: monitoring

- A way to keep track of a system's health and availability
- Common approach:
  - Watch for a trigger
  - Send an email alert to human
  - Wait for decision from human
- SRE approach:
  - Alerts
    - Human needs to take action immediately
  - Ticketing
    - Human need to take action, but not immediately
  - Logging
    - System handles the information

## Tenet 4: emergency response

- MTTF: mean time to failure
- MTTR: mean time to repair
- We can evaluate the effectiveness of emergency responses by considering how long it will take to bring the system back to health
- Note: human latency will play a role in this

# Tenet 5: change management

“SRE has found that roughly 70% of outages are due to changes in a live system. ”

– Benjamin Treynor Sloss, *Google SRE Book*

- SRE uses automation to:
  - Implement progressive rollouts
  - Detect problems (quickly and accurately)
  - Roll back changes when problems have been detected
- This minimizes the number of operations that are exposed to ‘bad’ changes
- Automation also removes humans from the loop (ish), which mitigates problems of:
  - Fatigue
  - Familiarity/contempt
  - Inattention to highly repetitive tasks

# Tenet 6: Capacity planning

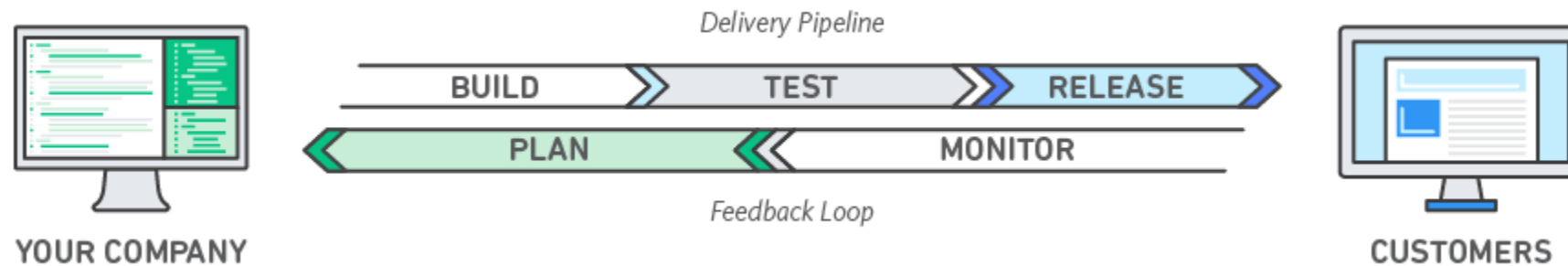
- Ensuring there is sufficient capacity and redundancy for future demand
- Ensuring that the capacity is in place by the time it is needed (before, not at or after)
- Should consider:
  - Organic growth: natural product adoption
  - Inorganic growth: growth stemming from feature launches, marketing, etc.
- Three capacity planning steps:
  1. Demand forecast for organic growth, extending beyond the lead time
  2. Incorporation of inorganic events
  3. Regular load testing
- Note, since SRE is in charge of capacity planning, they should also be in charge of provisioning

# Tenet 7 & 8: Efficiency and Performance

- Resource use involves
  - Demand (load)
  - Capacity
  - Efficiency
- SREs predict demand and capacity, and can modify the software/services for efficiency.
- SREs and developers should monitor and modify services to improve performance
  - i.e. Predict capacity
  - i.e. Add capacity (provisions)
  - i.e. Improve efficiency

Introducing DevOps/SRE on the cloud

# What does DevOps look like on the cloud?



# What does DevOps look like on the cloud?

- Speed
- Rapid delivery
- Reliability
- Scale

# What does DevOps look like on the cloud?

- Continuous integration
- Continuous delivery
- Micro-services
- Infrastructure as code
- Monitoring and logging

# What does DevOps look like on the cloud?

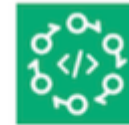
Cloud-providers provide us with DevOp services



Software Release Workflows  
AWS CodePipeline



Build and Test Code  
AWS CodeBuild



Unified CI/CD Projects  
AWS CodeStar



Deployment Automation  
AWS CodeDeploy

# What does DevOps look like on the cloud?

Cloud-providers provide us with DevOp services



Azure Pipelines



Azure Artifacts



Azure Test Plans



Azure Boards



Azure Repos

# What does DevOps look like on the cloud?

Cloud-providers provide us with DevOp services



Google Cloud Deploy



Cloud Build



Operations Suite



Artifact Registry



Binary Authorization

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